

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Sc. (AOC) Examination March 2018

OC 724: Separation Techniques

Date: 26/03/2018 Day: Monday Time: 10.00 a.m. to 01.00 p.m. Total Marks: 70

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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PART A Time: 10.00 a.m. to 10.30 a.m. Total marks: 20

Q-I A Choose the correct answer for the following questions. 10

1. In TGA, the following parameter is measured
 - A. ΔT vs. Temperature
 - B. ΔH vs. Temperature
 - C. ΔW vs. Temperature
 - D. A and B
2. In SCFC, the most suitable mobile phase is
 - A. CO
 - B. He
 - C. CO₂
 - D. A and B

3. In the normal phase chromatography, the stationary phase is
- A. Non-polar
 - B. Either non-polar or polar
 - C. Polar
 - D. None of the above
4. What useful information can be found from Van Deemter plot?
- A. K
 - B. Optimum column length
 - C. Optimum mobile flow rate
 - D. Longitudinal diffusion
5. A characteristic feature of any form of chromatography is use of
- A. An inert carrier gas
 - B. A mobile and a stationary phase
 - C. Water
 - D. Calculation of an R_f value
6. Which of the following is the most suitable gas to use as a carrier gas in GC.
- A. Chlorine
 - B. Oxygen
 - C. Helium
 - D. Methane
7. Which type of column has no eddy diffusion contribution to plate height?
- A. Polar LC columns
 - B. Non-polar LC columns
 - C. Wide bore packed columns
 - D. Capillary Columns
8. An electron capture detector is used
- A. In hollow cathodes lamps to regulate applied voltage
 - B. In photomultiplier tubes to measure current gain
 - C. For compounds which have elements with high electron affinities such as halogens
 - D. For radioactive compounds

9. A gradient elution in HPLC is one in which the composition of the solvent.
- A. Remain constant
 - B. Is changed continuously
 - C. A and B
 - D. None of the above
10. HPLC stands for
- A. High price Liquid Chromatography
 - B. High Problematic Chromatography
 - C. High promise Liquid Chromatography
 - D. High performance Liquid Chromatography

Q-I B Answer the following.

10

1. Write the equation of relative error due to separation and explain the term involved in it.

2. Mention the characteristics of solid support in GC.

3. Give the classification of chromatography based on mobile phase.

4. List the Characteristics required for mobile phase in GC.

5. Why Ghost Peaks (artifact) are observed in GC.

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OC 724: Separation Techniques

Date: 26/03/2018 Day: Monday Time: 10.30 a.m. to 01.00 p.m. Total Marks: 50

Important Instructions:

- Provide all necessary information as required in the answer sheet.
 - Attempt all questions.
 - Questions in **Part A** should be answered in the question paper itself.
 - Write answers for section I and section II of **Part B** in separate answer sheets.
 - Use of non-programmable calculators may be allowed if needed.
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PART B

SECTION-I

Q-II Answer the following.

- | | | |
|---|---|---|
| A | Mention the difference between GC and HPLC. | 4 |
| B | Give the characteristics of stationary phase in GC. | 2 |
| C | Explain the role of each term involved in Resolution. | 4 |

OR

- | | | |
|---|--------------------|---|
| C | Write note on FID. | 4 |
|---|--------------------|---|

Q-III Answer the following.

- | | | |
|---|--|---|
| A | Draw a block Diagram of HPLC. | 3 |
| B | Write the general requirement of mobile phase in HPLC. | 4 |
| C | Write note on plate theory. | 3 |

OR

- | | | |
|---|-------------------------------|---|
| C | Explain: NPC and RPC in HPLC. | 3 |
|---|-------------------------------|---|

SECTION-II

Q-IV Answer the following.

- | | | |
|---|---|---|
| A | Explain: Gradient analysis in HPLC. | 2 |
| B | Write note on: Refractive Index detector. | 4 |

C Give the basic detector requirement in case of HPLC. 4

OR

C Evaporative light scattering detector. 4

Q-V Answer the following.

A Write note on ECD. 3

B Give the method to determine C in H₂O₂. 3

C Write note on properties of supercritical fluid used in SCFC. 4

Q-VI Answer the following.

A Account for the formation of the fragments at $m/z = 70$ and $m/z = 42$ in the mass spectrum of 2-pentanol. Draw a block diagram of TGA, DTA, and DSC. 4

B What is the most characteristic fragmentation of aliphatic amines? 2

C Draw a block diagram of TGA, DTA, and DSC. 4

OR

C Mention the factor affecting Thermogravimetry. 4